U.S. Stock Market Linkages, UK Bond Yields, and FTSE 100 Returns

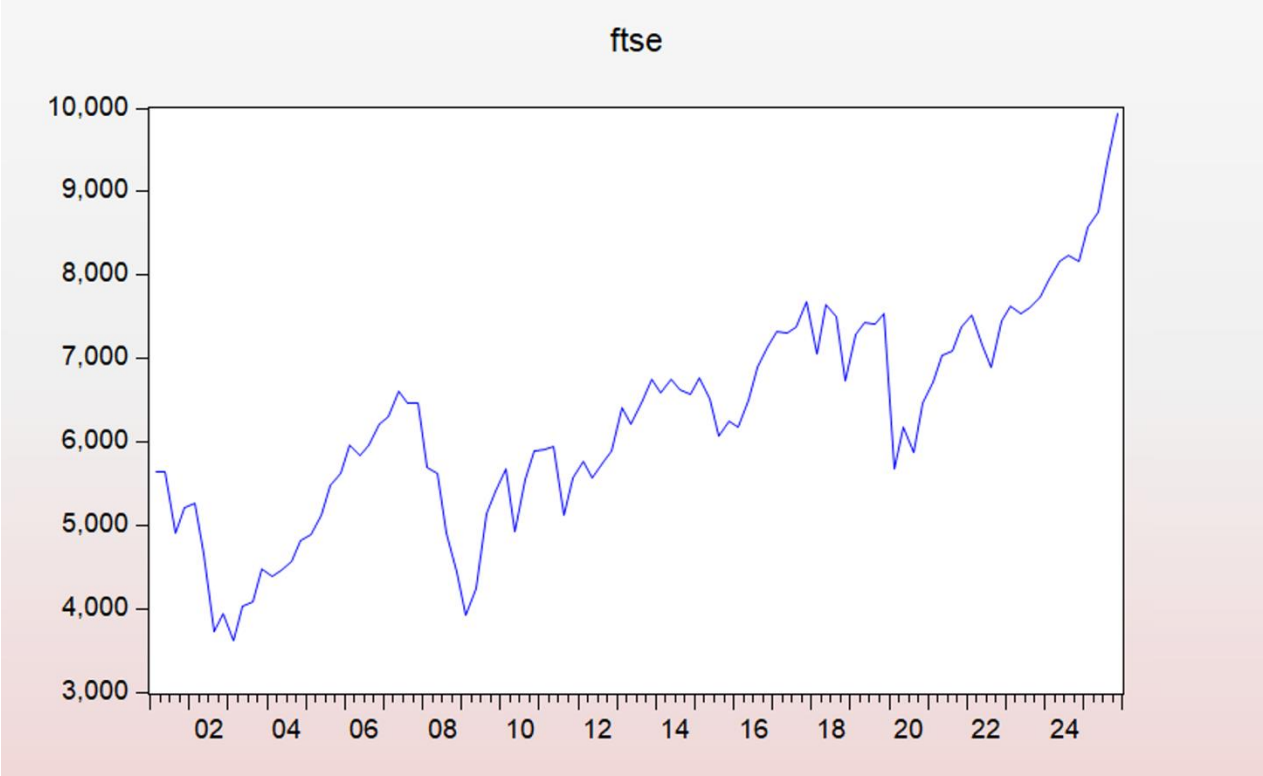


Research Question

To what extent are FTSE 100 quarterly returns explained by S&P 500 quarterly returns and changes in the UK 10-year government bond yield?

Project focus:	**FTSE_RET:** FTSE 100 quarterly return, UK stock market return
	SP500_RET: S&P 500 quarterly return, U.S. stock market return
	DUK10Y: Quarterly change in UK 10-year government bond yield

Background and Motivation



FTSE 100 Index Level, 2001Q1–2025Q4

- Wanted to better understand the UK economy and financial markets through real financial data
- Used the **FTSE 100** to represent the UK stock market and major UK-listed companies
- Used the **S&P 500** to test whether U.S. stock market movements are linked to FTSE 100 returns
- Included the **UK 10-year bond yield** to examine whether long-term interest-rate changes help explain stock returns

Literature Review

Stock Market Linkages and Stock-Bond Relationships

- Efficient market theory suggests stock prices respond quickly to available information, supporting the idea that U.S. market news may be reflected in UK stock returns (Fama, 1970)
- Eun and Shim find substantial multilateral interaction among national stock markets using a nine-market VAR model, supporting the idea that S&P 500 movements may help explain FTSE 100 returns (Eun & Shim, 1989)
- Hamao, Masulis, and Ng find short-run interdependence in price changes and volatility across New York, London, and Tokyo stock markets, which directly supports comparing S&P 500 and FTSE 100 returns (Hamao, Masulis, & Ng, 1990)
- Campbell and Ammer show that stock and 10-year bond returns are connected through news about future returns, inflation, and interest rates, supporting the inclusion of UK 10-year bond yield changes (Campbell & Ammer, 1993)
- Research gap: My project applies these ideas specifically to **FTSE 100 quarterly returns from 2001Q2–2025Q4**, testing whether UK returns are explained more by **S&P 500 returns** or **UK 10-year bond-yield changes**

Data Description

Data source: Investing.com
Frequency: Quarterly
Sample period: 2001Q1–2025Q4
Regression sample: 2001Q2–2025Q4
Observations: 99

Variable	Description	Transformation
ftse	FTSE 100 index level	Converted to ftse_ret
sp500	S&P 500 index level	Converted to sp500_ret
uk10y	UK 10-year bond yield	Converted to duk10y

Data 1: FTSE 100

- The **FTSE 100** is the dependent variable source
- It represents the UK stock market through large UK-listed companies
- Used the FTSE 100 index level to calculate **quarterly FTSE returns**
- Higher FTSE returns indicate stronger UK stock market performance

Data 2: S&P 500



The **S&P 500** represents the U.S. stock market



It is used as the main international stock-market linkage variable



Prior research shows major national stock markets are connected



Used S&P 500 index levels to calculate **quarterly S&P 500 returns**

Data 3: UK 10-Year Government Bond Yield



The UK 10-year bond yield reflects long-term interest-rate conditions



Bond yields may affect stock returns through discount rates and investor allocation between stocks and bonds



Rising bond yields can sometimes make bonds more attractive relative to stocks



I use the quarterly change in the UK 10-year yield instead of the level

Variables and Labels in Regression

Dependent Variable:

Variable	Label	Meaning
FTSE 100 return	FTSE_RET	UK stock market quarterly return

Independent Variables:

Variable	Label	Meaning
S&P 500 return	SP500_RET	U.S. stock market quarterly return
Change in UK 10Y yield	DUK10Y	Quarterly change in UK long-term bond yield

Testing Hypothesis

Hypothesis 1: U.S. stock-market linkage

- $H_0: \beta_1 = 0$
  S&P 500 returns do not significantly explain FTSE returns
- $H_1: \beta_1 \neq 0$
  S&P 500 returns significantly explain FTSE returns

Hypothesis 2: UK bond-yield effect

- $H_0: \beta_2 = 0$
  UK 10-year yield changes do not significantly explain FTSE returns
- $H_1: \beta_2 \neq 0$
  UK 10-year yield changes significantly explain FTSE returns

Expected signs:

- SP500_RET: positive
- DUK10Y: negative

Methodology



Model used: OLS regression with stationary quarterly data



Equation: $FTSE_RET_t = \beta_0 + \beta_1 SP500_RET_t + \beta_2 DUK10Y_t + u_t$



Tests used: ADF unit root tests, correlation matrix, OLS regression, residual correlogram, and ARCH test

	FTSE_RET	SP500_RET	DUK10Y
Mean	0.008260	0.021498	-0.003778
Median	0.020118	0.032542	-0.050000
Maximum	0.208201	0.199529	1.856000
Minimum	-0.247994	-0.225652	-1.439000
Std. Dev.	0.069738	0.081595	0.443756
Skewness	-0.923790	-0.848211	0.368704
Kurtosis	4.982422	3.711981	5.690559
Jarque-Bera	30.29215	13.96215	32.10438
Probability	0.000000	0.000929	0.000000
Sum	0.817783	2.128338	-0.374000
Sum Sq.	0.483365	0.698221	19.29949
Sum Sq. Dev.	0.476610	0.652466	19.29808
Observations	99	99	99

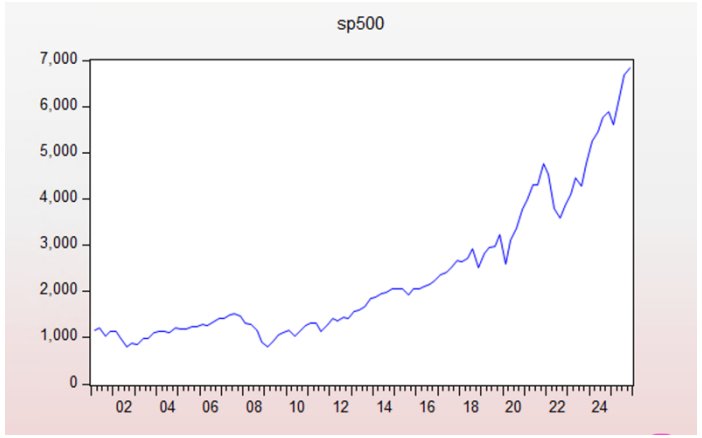
Descriptive Statistics

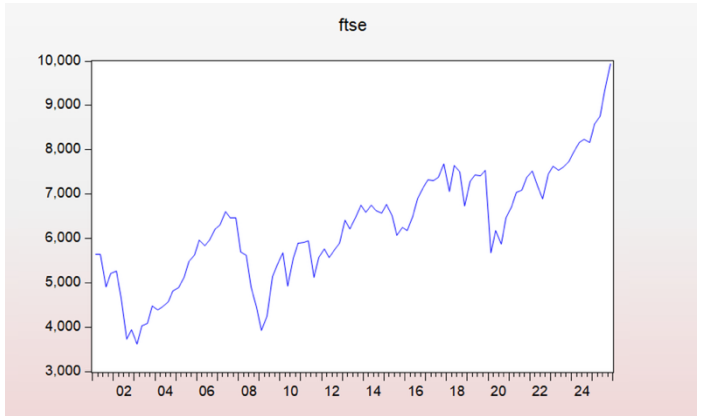
- FTSE average quarterly return ≈ **0.83%**
- S&P 500 average quarterly return ≈ **2.15%**
- S&P 500 returns show higher volatility than FTSE returns
- UK bond-yield changes are close to zero on average but have large movements in some periods

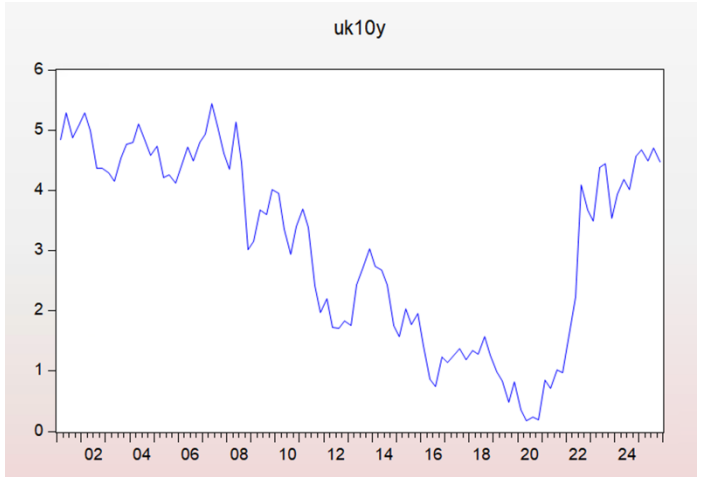
Correlation Results

	Correlation		
	SP500_RET	FTSE_RET	DUK10Y
SP500...	1.000000	0.832956	0.247079
FTSE_...	0.832956	1.000000	0.221094
DUK10Y	0.247079	0.221094	1.000000

- FTSE_RET and SP500_RET have a strong positive correlation (0.833)
- FTSE_RET and DUK10Y have a much weaker correlation (0.221)
- This suggests that U.S. stock-market returns are more closely related to FTSE 100 returns than UK bond-yield changes are
- This pattern is consistent with the regression results, where SP500_RET is significant but DUK10Y is not







Graphs & Trends

- FTSE level shows long-run movement over time
- UK10Y level declines for many years and rises sharply after 2021
- These visual patterns suggest the level variables may be nonstationary
- This motivates formal ADF unit root testing

ADF Unit Root Results

Variable	ADF Stat	p-value	Conclusion
FTSE	-0.3320	0.9151	Nonstationary
S&P 500	2.6193	1.0000	Nonstationary
UK10Y	-1.4927	0.5333	Nonstationary
FTSE_RET	-10.2165	0.0000	Stationary
SP500_RET	-9.6393	0.0000	Stationary
DUK10Y	-9.4306	0.0000	Stationary

Stationary Data Transformation

Transformations used:

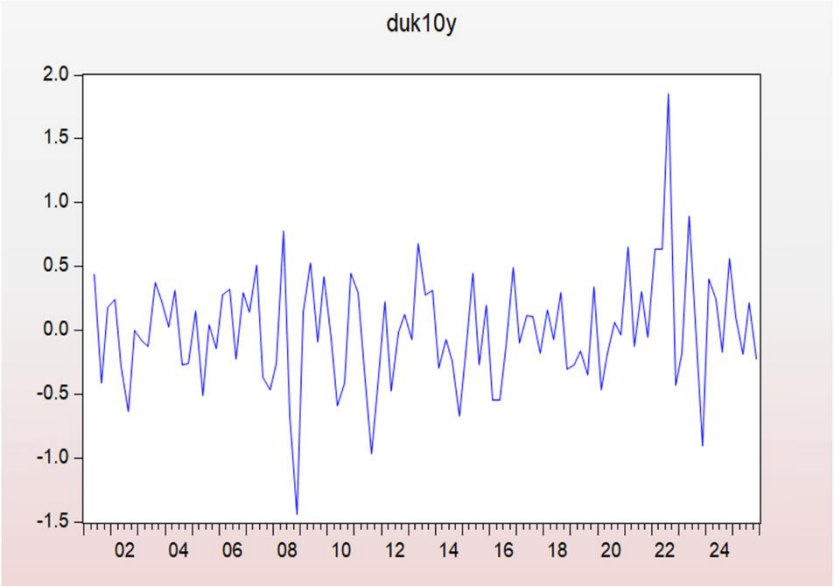
$$FTSE_RET_t = \frac{FTSE_t - FTSE_{t-1}}{FTSE_{t-1}}$$

$$SP500_RET_t = \frac{SP500_t - SP500_{t-1}}{SP500_{t-1}}$$

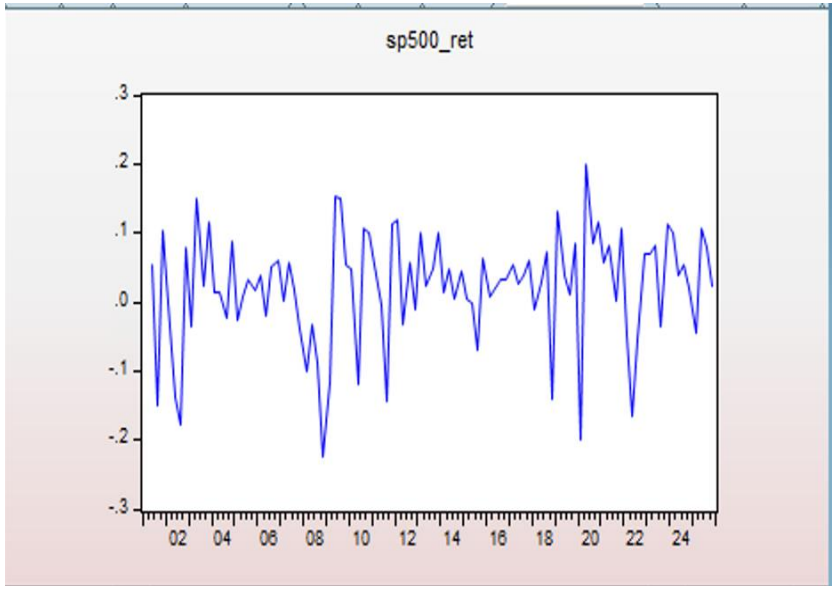
$$DUK10Y_t = UK10Y_t - UK10Y_{t-1}$$

Why transform the data?	Level variables were nonstationary
	Stationary data is required for the regression analysis
	Returns and yield changes are more appropriate for time-series regression

Transformed Graphs







- FTSE_RET fluctuates around zero
- DUK10Y fluctuates around zero
- The transformed variables visually appear more stationary
- Crisis periods create large movements, especially around 2008–2009 and 2020

OLS Regression Model

Dependent Variable: FTSE_RET
Method: Least Squares
Date: 05/08/26 Time: 01:14
Sample: 2001Q2 2025Q4
Included observations: 99

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.006961	0.004063	-1.713262	0.0899
SP500_RET	0.708471	0.049792	14.22862	0.0000
DUK10Y	0.002559	0.009155	0.279492	0.7805

R-squared	0.694065	Mean dependent var	0.008260
Adjusted R-squared	0.687692	S.D. dependent var	0.069738
S.E. of regression	0.038973	Akaike info criterion	-3.622079
Sum squared resid	0.145811	Schwarz criterion	-3.543439
Log likelihood	182.2929	Hannan-Quinn criter.	-3.590261
F-statistic	108.8963	Durbin-Watson stat	2.132581
Prob(F-statistic)	0.000000		

FTSE 100 quarterly returns are more strongly explained by U.S. stock market returns than by UK bond-yield changes

- S&P 500 returns are positive and statistically significant

  Coefficient = 0.7085, p-value = 0.0000

- UK 10-year bond-yield changes are not statistically significant

  Coefficient = 0.0026, p-value = 0.7805

- Model explains about 69.4% of FTSE return variation

  $R^2 = 0.6941$

Interpretation of Results

S&P 500 returns have a positive effect on FTSE 100 returns

- A 1 percentage point increase in S&P 500 quarterly returns is linked to a **0.71 percentage point increase** in FTSE 100 quarterly returns

Supports Hypothesis 1

- U.S. stock market movements help explain UK stock market returns

UK 10-year bond yield change is not significant

- High p-value means it does **not strongly explain** FTSE 100 returns in this model

Overall finding

- Results support international **stock-market linkage** literature more than stock-bond effect literature

Diagnostic Tests

- Residual correlogram p-values are above 0.05 → no major autocorrelation
- ARCH test p-value = 0.9200 → no ARCH effects
- GARCH/TGARCH is not needed for this quarterly model

Heteroskedasticity Test: ARCH

F-statistic	0.222773	Prob. F(4,90)	0.9251
Obs*R-squared	0.931375	Prob. Chi-Square(4)	0.9200

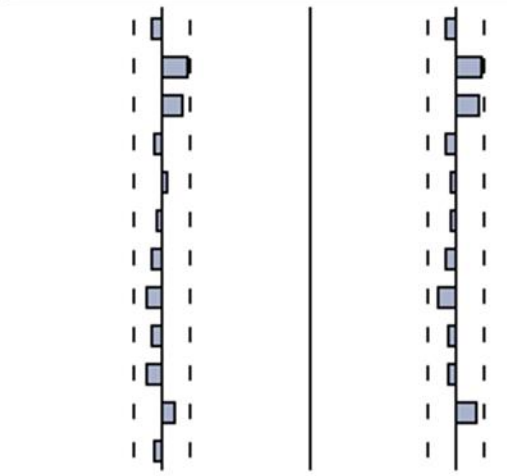
Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 05/08/26 Time: 01:48
Sample (adjusted): 2002Q2 2025Q4
Included observations: 95 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.001499	0.000377	3.971170	0.0001
RESID^2(-1)	0.037907	0.105426	0.359560	0.7200
RESID^2(-2)	0.065787	0.105443	0.623907	0.5343
RESID^2(-3)	-0.038592	0.105175	-0.366926	0.7145
RESID^2(-4)	-0.056464	0.110277	-0.512025	0.6099

R-squared	0.009804	Mean dependent var	0.001518
Adjusted R-squared	-0.034205	S.D. dependent var	0.002281
S.E. of regression	0.002320	Akaike info criterion	-9.243212
Sum squared resid	0.000484	Schwarz criterion	-9.108797
Log likelihood	444.0526	Hannan-Quinn criter.	-9.188898
F-statistic	0.222773	Durbin-Watson stat	2.016302
Prob(F-statistic)	0.925063		

ARCH Test

Sample: 2001Q2 2025Q4
Included observations: 99



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.079	-0.079	0.6431	0.423
		2	0.188	0.183	4.3038	0.116
		3	0.132	0.165	6.1212	0.106
		4	-0.062	-0.078	6.5268	0.163
		5	0.033	-0.038	6.6450	0.248
		6	-0.046	-0.042	6.8721	0.333
		7	-0.079	-0.071	7.5513	0.374
		8	-0.112	-0.122	8.9409	0.347
		9	-0.082	-0.064	9.6813	0.377
		10	-0.109	-0.065	11.025	0.356
		11	0.083	0.133	11.817	0.378
		12	-0.065	-0.004	12.305	0.421

Residual correlogram

Findings & Conclusion

SP500_RET has a strong, positive, and statistically significant effect on FTSE_RET

DUK10Y does not have a statistically significant effect on FTSE_RET

The model explains about **69.4%** of FTSE return variation

The results suggest that FTSE 100 quarterly returns are more connected to **U.S. stock market movements** than to **UK bond-yield changes**

My findings support the literature on international stock-market linkages

Limitations

The model shows association, not causation

Quarterly data may hide short-run market reactions

Only two explanatory variables are included

Other factors may matter, such as exchange rates, UK inflation, oil prices, Bank of England policy, Brexit, COVID-19, and global risk

Crisis events are not separately modeled

GARCH/TGARCH was not necessary here, but monthly or daily data may show stronger volatility clustering

Next Steps

Completed in Part 2

- Switched from quarterly to monthly data for more observations
- Added the GBP/USD exchange rate as a third explanatory variable
- Used a VAR model to study dynamic relationships among FTSE_RET, SP500_RET, DUK10Y and GBPUSD_RET
- Ran Granger causality tests to check whether past S&P 500 returns help forecast FTSE returns

For future research

- Add event dummy variables for major shocks:
  - 2008 financial crisis
  - Brexit referendum period
  - COVID-19 shock
  - 2022 rate-hike period
- Add UK inflation and the Bank of England policy rate
- Test whether volatility effects are stronger at a higher frequency

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